

• • LIVE WEEKEND COHORT

Maths for ML — Power Weekend

Calculus + Linear Algebra + Probability in one focused weekend.

The complete mathematical toolkit for machine learning, taught for people who want to understand AI deeply — not just use it.

PRICE	DURATION	LEVEL	FORMAT	SCHEDULE
₹2,499	2 Days · 14 hrs	Intermediate	Live	Bi-weekly weekends



Prasant Mishra

Founder & Instructor, Qurious Academy

What is Maths for ML — Power Weekend?

Machine learning is built on three mathematical pillars: calculus (for optimisation and backpropagation), linear algebra (for data representation and transformation), and probability & statistics (for reasoning under uncertainty and model evaluation). Most practitioners lack one or more of these, and that gap slowly limits what they can do.

This weekend cohort covers all three — deeply and clearly — in two focused days. It's designed for people who are already working with ML or AI and want to go deeper, or who are preparing for the AI & ML Engineering 12-Week program and want a solid mathematical foundation before it begins.

The teaching is visual and intuition-first throughout. Proofs and abstract formalism are deferred in favour of geometric and probabilistic understanding. By Sunday afternoon you'll be able to read the maths in ML papers without fear, understand why gradient descent converges, explain what PCA is doing, and solve probability problems of the kind that appear in ML interviews. A problem-solving marathon and personal study plan close out Day 2.

IS THIS RIGHT FOR YOU?

Who Should Attend

✓ THIS IS FOR YOU IF...

- ✓ You're working in ML/AI and want to understand the maths rather than treat it as a black box
- ✓ You're preparing to enroll in the 12-Week AI & ML Engineering program and want to arrive with strong mathematical foundations
- ✓ You want to pass ML engineering interviews that test statistics, probability, and linear algebra
- ✓ You have 12th-grade maths and want a fast, focused path to ML-relevant mathematics without taking a full university course

✗ THIS IS NOT FOR YOU IF...

- ✗ You have no maths background at all — you need 12th-grade algebra, basic trigonometry, and some exposure to functions
- ✗ You want pure mathematics (rigorous proofs, abstract algebra, analysis) — this is applied mathematics for ML practitioners
- ✗ You want just one area covered deeply — the individual masterclasses (Calculus, Linear Algebra) give more time to each topic separately

Prerequisites: 12th-grade mathematics — algebra, functions, basic trigonometry, coordinate geometry. No calculus or linear algebra experience required. Basic Python is helpful for the NumPy session but not mandatory.

Two Days, Four Blocks

DAY 1 — MORNING**Calculus for Optimisation**

- ▶ **Derivatives & partial derivatives** — single-variable derivatives reviewed, then extended to multivariable functions; the partial derivative as "rate of change while holding others constant"
- ▶ **Gradient descent from scratch** — implementing gradient descent step by step; the gradient vector as the direction of steepest ascent and why we go in the opposite direction
- ▶ **Chain rule & backpropagation intuition** — the chain rule as the mechanism behind backpropagation; how gradients flow backwards through a computational graph
- ▶ **Multivariable optimisation** — finding minima of functions of multiple variables; saddle points, local vs. global minima, and what this means for training neural networks

DAY 1 — AFTERNOON**Linear Algebra**

- ▶ **Vectors, matrices & transformations** — vectors as geometric objects, matrices as transformations; the visual intuition that makes ML notation readable
- ▶ **Matrix multiplication** — as composition of transformations; dot products, inner products, and why matrix operations are the language of neural networks
- ▶ **Eigenvalues & PCA** — eigendecomposition explained geometrically; how PCA uses eigenvectors to find the directions of maximum variance in data
- ▶ **NumPy hands-on** — implementing matrix operations, eigendecomposition, and PCA in NumPy; tying the session's maths to real ML code

DAY 2 — MORNING**Probability & Statistics**

- ▶ **Probability basics** — sample spaces, events, conditional probability, independence; common distributions (Gaussian, Bernoulli, Multinomial) and where they appear in ML
- ▶ **Expected value & variance** — the mathematical definition of "average" and "spread"; why variance matters for model generalisation and regularisation
- ▶ **MLE & MAP estimation** — Maximum Likelihood Estimation as the foundation of most ML training; the Bayesian extension via Maximum A Posteriori; how this connects to loss functions
- ▶ **Information theory basics** — entropy as "uncertainty", KL divergence as "difference between distributions"; why these appear in VAEs, GANs, and language model training

DAY 2 — AFTERNOON**Tying It Together**

- ▶ **How all three areas combine in a neural network** — a complete walkthrough of the mathematical story of a training step: linear algebra (forward pass), calculus (gradient computation), probability (loss function)
- ▶ **Reading an ML paper's maths section** — a live session working through a real paper excerpt; applying what you've learned to actual research notation

- ▶ **Problem-solving marathon** — a structured set of problems spanning all three areas, worked collaboratively; exam-style and interview-style questions
- ▶ **Personal study plan** — based on your goals and which areas need more depth, a concrete continuation plan for going further in each subject

LEARNING OUTCOMES

What You'll Walk Away With

01 Understand gradient descent mathematically

— not just "it minimises the loss" but the full mathematical story of how partial derivatives guide parameter updates

02 Read ML papers without fear

— the mathematical notation in research papers becomes navigable once you have geometric and probabilistic intuition behind the symbols

03 Use NumPy for matrix operations

— translate mathematical expressions directly into NumPy code; the practical foundation for every ML framework

04 Solve probability problems for ML interviews

— Bayes' theorem, distributions, MLE, and information theory — the statistical questions that appear in ML engineering interviews

WHAT'S INCLUDED

Everything You Get

✓ **Both day recordings** — full HD recordings of both days shared within 24 hours

✓ **Problem sets with full solutions** — a comprehensive set of problems covering all three mathematical areas, with complete worked solutions

✓ **Formula reference PDF** — a clean, organised reference sheet for all key formulae from calculus, linear algebra, and probability

✓ **Learner's Certificate** — digital certificate of completion from Qurious Academy

LOGISTICS

Schedule & Pricing

SCHEDULE

Every other weekend (bi-weekly)

TIME

9:00 AM – 4:00 PM IST

DURATION

2 days · ~14 hours total

PRICE

₹2,499

❖ **How does this compare to taking the three individual masterclasses?**

The Power Weekend is more cohesive — it explicitly connects all three areas and ends with a unifying session on how they combine in a neural network. The individual masterclasses give more time to each topic. Both paths are valid.

❖ **Is this a prerequisite for the 12-Week AI & ML Engineering program?**

No — the 12-Week program introduces the relevant maths as needed. However, attending the Power Weekend first means you arrive with stronger foundations and can focus on the ML applications rather than catching up on maths.

❖ **Do I need a calculator or any specific tools?**

A scientific calculator is helpful for the probability section. For the NumPy session, you'll need a laptop with Python and NumPy installed — we send setup instructions beforehand.

About Prasant Mishra



Prasant Mishra

Founder & Instructor — Qurious Academy

GenAI Solutions Architect

15+ Yrs Teaching

2500+ Students

60+ Projects Delivered

Prasant brings the unusual combination of deep mathematical knowledge and daily applied AI work. He's taught calculus, linear algebra, and probability to thousands of students — and he uses all three constantly in his work designing AI systems. That dual perspective lets him be precise about what actually matters for ML practice vs. what can be safely deferred.

The Power Weekend reflects his view that the three areas of ML mathematics aren't separate subjects — they're three lenses on the same underlying ideas, and understanding how they connect is as important as understanding each one individually.

QuriousAcademy

Focused, live learning for curious minds. No fluff — just the concepts that matter.

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Live Cohort · Bi-weekly weekends · 9 AM–4 PM IST

₹2,499

Enroll at quriousacademy.com

or visit quriousacademy.com/courses

Questions? Reach us at hello@quriousacademy.com

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